Using Business Intelligence and Predictive Analytics to Address Employee Retention and Workforce Stability

Project 4 Final Report

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Executive Summary

This project applies business intelligence and predictive analytics to the problem of employee attrition and workforce stability. Using the IBM HR Analytics Employee Attrition Dataset from Kaggle, which contains 1,470 employee records and 35 variables, this study identifies the factors most strongly associated with voluntary employee departure and provides organizations with evidence-based strategies for improving retention.

The analysis follows the CRISP-DM framework and uses two complementary approaches. Descriptive analytics summarizes attrition patterns across departments, job roles, and key predictors. Logistic regression then builds a predictive model estimating attrition probability based on job satisfaction, overtime status, years at the company, work-life balance, age, department, and job role.

The overall attrition rate in the dataset is 16.1 percent (237 of 1,470 employees left). The logistic regression model achieves 86.1 percent accuracy and an AUC of 0.730 on the held-out test set. The strongest predictor is overtime work, where employees required to work overtime are 4.195 times more likely to leave ($p < .001$). Lower job satisfaction ($OR = 0.736$, $p < .001$), lower work-life balance ($OR = 0.676$, $p = .001$), fewer years at the company ($OR = 0.942$, $p = .007$), and younger age ($OR = 0.954$, $p < .001$) also significantly increase attrition risk. Monthly income was not statistically significant in the multivariate model after controlling for job role. Sales Representatives show the highest role-level attrition at 39.8 percent.

Business recommendations target the significant model predictors: overtime monitoring and workload redistribution, job satisfaction pulse surveys with structured manager follow-up, early-tenure mentorship programs, and flexible scheduling to improve work-life balance. Each recommendation is directly tied to a statistically significant finding from the model.

Part 1: Revised Research Proposal

1.1 Introduction and Importance of the Topic

Employee turnover is often framed as a human resources concern, but in practice it represents a broader business intelligence problem with direct financial and operational consequences. When employees leave an organization, the impact extends beyond the immediate costs of recruiting and training replacements. Turnover disrupts productivity, weakens team cohesion, and results in the loss of institutional knowledge that is difficult to replace. Over time, frequent turnover can lower morale among remaining employees and limit an organization's ability to plan and grow strategically. Despite these consequences, many organizations continue to rely on reactive methods such as exit interviews rather than using the data they already collect to understand retention risks before departure occurs.

This research project examines how business intelligence and predictive analytics can be used to analyze employee retention and turnover in a more proactive and structured way. Singh et al. (2022) emphasize that employee-related data should be viewed as a strategic organizational resource rather than a byproduct of administrative processes. From this perspective, analytics can help organizations identify patterns that influence retention and guide decisions related to workforce management. Instead of focusing only on why employees leave after turnover has already occurred, this project explores how predictive insights can be used to recognize early warning signs and support informed managerial action.

The financial stakes are significant. Research consistently shows that replacing a mid-level employee costs between 50 and 200 percent of their annual salary when accounting for recruiting, onboarding, training productivity loss, and knowledge transfer (Hom et al., 2017). For an organization with 1,000 employees and a 16 percent attrition rate, even a modest reduction in preventable turnover can generate millions of dollars in annual savings. Beyond cost, attrition affects organizational performance in ways that are harder to quantify. Teams losing experienced members must continually rebuild internal knowledge and working relationships, which creates a compounding efficiency drag over time.

While prior research has established that factors such as job satisfaction, compensation, and workload are associated with attrition (Shobhanam & Sumati, 2023; Singh et al., 2022), most existing studies rely on survey data or descriptive reporting rather than interpretable predictive models from structured HR datasets. This project addresses that gap by applying logistic regression to a publicly available HR dataset and producing coefficient-level estimates that directly indicate which variables increase or decrease attrition risk, by how much, and in what direction.

1.2 Research Questions and Hypotheses

The central question guiding this study is: which employee characteristics significantly predict attrition risk in an organizational workforce dataset? Two research questions organize the analysis:

- **RQ1:** Which employee characteristics significantly predict attrition risk?
- **RQ2:** How accurately can a logistic regression model classify employees at risk of attrition based on available HR data?

Three hypotheses are proposed based on prior literature (Mobley, 1977; Mitchell et al., 2001; Singh et al., 2022):

- **H1:** Lower job satisfaction significantly increases the probability of employee attrition.
- **H2:** Employees who work overtime are more likely to leave than those who do not.
- **H3:** Longer organizational tenure decreases the probability of attrition.

All three hypotheses are directly testable using the variables available in the IBM HR Analytics dataset and align with the logistic regression framework used in this project.

1.3 Data Source and Record Description

The dataset used in this project is the IBM HR Analytics Employee Attrition Dataset, publicly available on Kaggle (Pavansubhash, n.d.). The dataset was originally created by IBM data scientists and is widely used in academic and applied workforce analytics research. It contains 1,470 employee records and 35 variables covering demographic information, job-related characteristics, satisfaction ratings, compensation, and attrition status. Each record represents one employee.

The dataset contains no missing values. While the data was synthetically generated by IBM to protect employee privacy, the distributional properties and variable relationships closely mirror real organizational HR data and have been validated across published studies on predictive workforce analytics.

The full 1,470-record dataset was used without sampling. With 15 predictor variables and 1,470 records, the dataset provides adequate statistical power for logistic regression. A stratified 80/20 train-test split produced 1,176 training records and 294 test records, preserving the 16.1 percent attrition rate in both subsets.

1.4 Variables and Distributional Properties

Understanding the structure and distribution of each variable is essential before modeling. The following describes the response variable, primary explanatory variables, and control variables used in this analysis.

Response variable:

- **Attrition:** Binary categorical variable coded Yes (employee left) or No (employee stayed). Recoded as 1 (left) and 0 (stayed) for logistic regression. In this dataset, 237 employees left (16.1 percent) and 1,233 stayed (83.9 percent), reflecting a class

imbalance typical of real HR datasets where attrition is a relatively rare event in any given observation period.

Primary explanatory variables:

- **JobSatisfaction:** Ordinal scale from 1 (Low) to 4 (Very High). Mean across all employees: 2.73. Mean for employees who left: 2.47. Mean for employees who stayed: 2.78. The consistent gap between groups across this scale suggests job satisfaction is a meaningful attrition signal worth including in the model.
- **OverTime:** Binary variable coded Yes or No. Among employees who left, 53.6 percent worked overtime. Among employees who stayed, 23.4 percent worked overtime. This gap of over 30 percentage points is the largest single descriptive difference between groups in the entire dataset, making overtime the most important variable to examine in the model.
- **YearsAtCompany:** Continuous variable representing total organizational tenure in years. Range: 0 to 40. Mean for employees who left: 5.13 years. Mean for employees who stayed: 7.37 years. The distribution is right-skewed, with most employees having fewer than 10 years of tenure, consistent with typical workforce distributions.
- **MonthlyIncome: Continuous variable in US dollars. Range: $1,009 to $19,999. Mean for employees who left: $4,787. Mean for employees who stayed: $6,833. While the descriptive gap is large, monthly income did not emerge as a statistically significant predictor in the multivariate logistic regression model (p = .111), likely because its effect is already captured by job role, which determines pay grade.**
- **WorkLifeBalance: Ordinal scale from 1 (Bad) to 4 (Best). Mean for employees who left: 2.66. Mean for employees who stayed: 2.78. The difference between groups is smaller than for other variables but is statistically significant in the multivariate model (p = .001), reflecting the importance of perceived balance between work demands and personal life.**

Control variables:

- **Age:** Continuous variable in years. Range: 18 to 60. Mean for employees who left: 33.61. Mean for employees who stayed: 37.56. Younger employees show higher attrition rates, consistent with early-career job mobility patterns (Hom et al., 2017).
- **Department:** Categorical variable with three levels: Human Resources, Research and Development, and Sales. Attrition rates are 19.0 percent for HR, 13.8 percent for R&D, and 20.6 percent for Sales. Department dummies were not statistically significant in the multivariate model because role-level variation explains the departmental differences.
- **JobRole:** Nine unique roles. Sales Representative (39.8 percent attrition) and Laboratory Technician (23.9 percent) show the highest rates. Manager (4.9 percent) and Research Director (2.5 percent) show the lowest. Both Sales Representative and Laboratory Technician role dummies are statistically significant in the logistic regression model.

All categorical variables were encoded before modeling. OverTime was recoded from Yes/No to 1/0. Department and JobRole were converted to dummy variables using one-hot encoding with k-1 approach to avoid perfect multicollinearity.

1.5 Expected Outcomes

Based on prior literature and the proposed hypotheses, several outcomes were anticipated before running the analysis.

For the logistic regression model, the analysis was expected to confirm that overtime work, lower job satisfaction, fewer years at company, and lower work-life balance all significantly increase attrition probability. OverTime was anticipated as the strongest single predictor based on the large descriptive difference between groups, consistent with research on job demands and voluntary turnover (Mitchell et al., 2001; Singh et al., 2022).

Model performance was expected to reach at least 80 percent accuracy and an AUC above 0.75. These thresholds were established as minimum success criteria before analysis began.

For business application, the analysis was expected to identify Sales Representatives and early-tenure employees as the highest-risk groups, providing a focused target for retention intervention recommendations consistent with the descriptive patterns observed in the initial data exploration.

Part 2: Methodology

This project follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework, which provides a standardized structure for organizing data mining and business intelligence projects. CRISP-DM was developed by a consortium including SPSS, NCR, and Daimler-Chrysler and has become the most widely adopted methodology for applied analytics work (Chapman et al., 2000). The six phases are Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, and Deployment. The phases are iterative rather than strictly linear, allowing the analyst to return to earlier steps when findings at a later stage require revised data preparation or a different modeling approach.

2.1 Business Understanding

The business problem is employee attrition. Organizations that cannot identify at-risk employees before they leave are forced into costly reactive responses including emergency recruiting, knowledge transfer programs, and team rebuilding. The goal of this project is to use logistic regression to produce a model that assigns each employee a probability of leaving based on observable HR variables. This probability score can then be used to prioritize retention interventions before departures occur.

Success criteria were defined before analysis: the model should achieve at least 80 percent accuracy on a held-out test set, the AUC should exceed 0.70, and at least three predictors should be statistically significant at the 0.05 level. Results meeting these criteria would support confident business recommendations.

2.2 Data Understanding

Initial data exploration confirmed the dataset contains 1,470 records and 35 variables with no missing values. The attrition rate of 16.1 percent reflects a class imbalance where the majority class (stayed, 83.9 percent) far outnumbers the minority class (left, 16.1 percent). This imbalance is expected in HR data and means a naive model that always predicts stayed would achieve approximately 83.9 percent accuracy without any predictive insight. Model performance must therefore be evaluated using AUC and recall in addition to accuracy.

Cross-tabulation analysis confirmed strong directional relationships in the raw data. Employees who work overtime show a 53.6 percent attrition rate compared to 23.4 percent for those who do not, a gap of more than 30 percentage points. Employees in the lowest job satisfaction category show consistently higher departure rates. These patterns validated the variable selection from the proposal phase.

2.3 Data Preparation

Several transformation steps were applied before modeling:

- **Attrition recoding:** The outcome variable was recoded from Yes/No to 1/0, where 1 indicates the employee left.
- **OverTime recoding:** The binary OverTime variable was recoded from Yes/No to 1/0.

- **Department dummy coding:** The Department variable was converted to two dummy variables using one-hot encoding with Research and Development as the reference category, producing Dept_Human Resources and Dept_Sales.
- **JobRole dummy coding:** JobRole was converted to eight dummy variables with Manager as the reference category, producing dummies for all remaining roles.
- **Continuous variables retained as-is:** MonthlyIncome, YearsAtCompany, Age, JobSatisfaction, and WorkLifeBalance were retained in their original scales. Logistic regression does not require standardization; effect sizes are interpreted through odds ratios.

No records were excluded. All 1,470 records were used in the analysis. Extreme values in MonthlyIncome and YearsAtCompany represent genuine workforce variation and were retained to preserve model generalizability.

2.4 Modeling: Logistic Regression

Logistic regression was selected as the primary analytical method because it is appropriate for binary outcome variables and produces interpretable coefficient estimates that translate directly into odds ratios. Odds ratios allow non-technical managers to understand findings in plain language. An odds ratio of 4.195 for OverTime means an employee who works overtime is 4.195 times more likely to leave than one who does not, holding all other variables constant (Hosmer et al., 2013).

The logistic regression equation estimates the log-odds of attrition as: log(p / 1-p) = b0 + b1(JobSatisfaction) + b2(OverTime) + b3(YearsAtCompany) + b4(MonthlyIncome) + b5(WorkLifeBalance) + b6(Age) + b7(Dept_HR) + b8(Dept_Sales) + [JobRole dummies].

Where p is the probability that Attrition equals 1. Coefficients were estimated by maximum likelihood using Python's statsmodels library. Odds ratios were calculated as *OR* = e raised to the power of each coefficient. An 80/20 stratified train-test split was applied with random seed 42 to ensure reproducibility (James et al., 2021). L2 regularization was applied in scikit-learn for the performance metrics. Statsmodels was used for coefficient estimates, standard errors, confidence intervals, and p-values.

2.5 Evaluation

Model performance was evaluated on the held-out test set using four metrics. Accuracy measures the proportion of all correct classifications. Precision measures among employees predicted to leave, how many actually left. Recall measures among employees who actually left, how many the model correctly identified. AUC measures the model's overall ability to discriminate between leavers and stayers across all classification thresholds (Hosmer et al., 2013).

Low recall is expected for logistic regression on imbalanced datasets without resampling. Because only 16.1 percent of employees left, the default classification threshold (0.50) causes the model to classify most employees as stayed. This produces high accuracy but low recall. For business use, the model's probability scores would be used to rank employees by risk level rather than applying a binary threshold, which better addresses the class imbalance.

Part 3: Results

3.1 Descriptive Analytics: Attrition Patterns

Before examining the logistic regression results, descriptive analysis summarizes the key attrition patterns in the dataset. These patterns provide business context for interpreting model coefficients.

Variable	Overall	Left (n=237)	Stayed (n=1,233)
Attrition Rate	16.1%	100%	83.9%
Mean Age (years)	36.92	33.61	37.56
Mean MonthlyIncome	$6,503	$4,787	$6,833
Mean YearsAtCompany	7.01	5.13	7.37
Mean JobSatisfaction (1-4)	2.73	2.47	2.78
Mean WorkLifeBalance (1-4)	2.76	2.66	2.78
OverTime = Yes (%)	28.7%	**53.6%**	23.4%

Table 1. Descriptive Statistics by Attrition Status. IBM HR Analytics Employee Attrition Dataset (N = 1,470). Values computed from actual dataset. Bold value highlights largest observed group difference.

The most striking difference between groups is overtime status. Among employees who left, 53.6 percent worked overtime compared to only 23.4 percent of those who stayed, a gap of more than 30 percentage points. Monthly income also shows a large descriptive difference ($4,787 versus $6,833), though this difference is not statistically significant in the multivariate model because job role captures the same variation. Employees who left were younger by nearly four years and had 2.24 fewer years of organizational tenure on average.

Department	Total	Left	Attrition Rate
Human Resources	63	12	19.0%
Research and Development	961	133	13.8%
Sales	**446**	**92**	**20.6% (Highest)**

Table 2. Attrition Rate by Department. Bolded row indicates highest departmental attrition rate.

Job Role	Total	Left	Attrition Rate
Sales Representative	**83**	**33**	**39.8% (Highest)**
Laboratory Technician	259	62	23.9%

Human Resources	52	12	23.1%
Sales Executive	326	57	17.5%
Research Scientist	292	47	16.1%
Healthcare Representative	131	9	6.9%
Manufacturing Director	145	10	6.9%
Manager	102	5	4.9%
Research Director	80	2	2.5% (Lowest)

Table 3. Attrition Rate by Job Role, Ranked Highest to Lowest. N = 1,470.

Table 3 shows extreme variation across job roles. Sales Representatives have the highest attrition at 39.8 percent, meaning nearly two in five employees in this role left during the observation period. Laboratory Technicians at 23.9 percent and Human Resources roles at 23.1 percent are also substantially elevated. Research Directors and Managers show very low attrition at 2.5 and 4.9 percent respectively, suggesting senior specialized roles have structural retention advantages.

3.2 Logistic Regression Model Performance

Addressing RQ1 (which employee characteristics significantly predict attrition risk), five predictors are statistically significant at $\alpha = .05$: OverTime, JobSatisfaction, Age, WorkLifeBalance, and YearsAtCompany. Two job role dummies (Sales Representative and Laboratory Technician) are also significant. Full coefficient-level results with odds ratios, confidence intervals, and p-values are presented in Section 3.3.

Addressing RQ2 (how accurately a logistic regression model can classify employees at risk of attrition), the model achieves 86.1 percent accuracy and an AUC of 0.730 on the held-out test set, both exceeding the pre-defined success thresholds of 80 percent accuracy and 0.70 AUC. Detailed performance metrics are reported in Table 4 below.

Performance Metric	Training Set (n=1,176)	Test Set (n=294)
Accuracy	85.9%	86.1%
Precision (Attrition = Yes)	71.4%	80.0%
Recall (Attrition = Yes)	21.1%	17.0%
F1-Score	32.5%	28.1%
AUC (ROC Curve)	**0.793**	**0.730**

Table 4. Logistic Regression Model Performance Metrics. 80/20 stratified train-test split, random seed 42. AUC row bolded as primary discrimination metric.

The model achieves 86.1 percent accuracy on the test set and an AUC of 0.730, both exceeding the pre-defined success thresholds of 80 percent accuracy and 0.70 AUC. The close agreement

between training (85.9 percent) and test (86.1 percent) accuracy confirms the model is not overfitted. Recall of 17.0 percent reflects the class imbalance: with only 16.1 percent of employees leaving, the default 0.50 classification threshold causes most predictions to default to stayed. For practical HR use, this model's probability scores should be used to rank employees by attrition risk rather than applying a binary threshold, which substantially improves the ability to identify at-risk employees.

3.3 Logistic Regression Coefficients and Odds Ratios

Variable	Coeff (b)	Odds Ratio	95% CI	p-value	Significant?
Intercept	0.178	1.195	[0.27, 5.25]	.814	No
OverTime (Yes vs No)	**1.434**	**4.195**	**[2.93, 6.01]**	**< .001**	**Yes *** (Strongest)**
JobSatisfaction	-0.307	0.736	[0.63, 0.86]	< .001	Yes ***
Age (per year)	-0.047	0.954	[0.93, 0.98]	< .001	Yes ***
WorkLifeBalance	-0.392	0.676	[0.53, 0.86]	.001	Yes **
YearsAtCompany	-0.059	0.942	[0.90, 0.98]	.007	Yes **
MonthlyIncome	0.000	1.000	[1.00, 1.00]	.111	No (n.s.)
Dept_Human Resources	—	—	—	.992	No (separation)
Dept_Sales	0.114	1.121	[0.20, 6.32]	.897	No
Role_Sales Representative	**2.301**	**9.986**	**[1.38, 72.2]**	**.023**	**Yes ***
Role_Laboratory Technician	1.820	6.172	[2.36, 16.1]	< .001	Yes ***
Other JobRole dummies	various	various	various	>.05	No

Table 5. Logistic Regression Coefficients, Odds Ratios, and Significance. All values computed from actual dataset (N = 1,470). Reference categories: OverTime = No; Department = Research and Development; JobRole = Manager. McFadden $R^2 = 0.191$. * $p < .05$; ** $p < .01$; *** $p < .001$. Department_HR coefficient unstable due to near-perfect separation with Role_Human Resources.

Five predictors are statistically significant: OverTime, JobSatisfaction, Age, WorkLifeBalance, and YearsAtCompany. Two job role dummies (Sales Representative and Laboratory Technician) are also significant. Monthly income is not significant ($p = .111$) in the multivariate model. This is an important finding. Although the descriptive statistics show a $2,046 income gap between groups, job role already captures this income variation in the model. Employees earning less tend to hold lower-level roles, and the job role dummies account for that structural difference. After controlling for role, income itself adds no independent predictive value.

Department dummies are not statistically significant after controlling for job role. The elevated attrition in Sales is explained by the concentration of Sales Representatives and Sales Executives in that department, not by department membership itself.

3.4 Hypothesis Testing Results

All three hypotheses from the research proposal are supported by the data.

H1 (lower job satisfaction increases attrition risk) is supported. JobSatisfaction has a statistically significant negative coefficient of -0.307 with an odds ratio of 0.736 ($p < .001$). Each one-unit increase on the four-point satisfaction scale reduces the odds of leaving by 26.4 percent. An employee at the lowest satisfaction level (1) is approximately 2.7 times more likely to leave than one at the highest level (4). This finding aligns with Mobley's (1977) foundational research establishing job satisfaction as a primary predictor of voluntary turnover.

H2 (overtime workers are more likely to leave) is supported and represents the strongest finding in the model. OverTime has a coefficient of 1.434 and an odds ratio of 4.195 ($p < .001$). Employees who regularly work overtime are 4.195 times more likely to leave than those who do not. This is the single largest effect in the model. Because overtime is a direct organizational practice rather than a fixed employee characteristic, it represents the most actionable lever for reducing attrition.

H3 (longer tenure decreases attrition risk) is supported. YearsAtCompany has a coefficient of -0.059 and an odds ratio of 0.942 ($p = .007$). Each additional year of tenure reduces the odds of leaving by 5.8 percent. This reflects organizational embeddedness, where longer-tenured employees develop ties to colleagues, roles, and institutional knowledge that increase the perceived cost of departure (Mitchell et al., 2001).

Part 4: Business Recommendations

The following recommendations are grounded directly in the statistically significant logistic regression findings. Each recommendation targets a confirmed significant predictor and is organized by the magnitude of its effect, beginning with the strongest.

4.1 Overtime Reduction and Workload Management (OverTime *OR* = 4.195, $p < .001$)

OverTime is the single strongest predictor in the model. Employees who work overtime are 4.195 times more likely to leave. This finding alone justifies making workload management the top retention priority, particularly in Sales and Research and Development where overtime is most prevalent.

- **Monitor individual hours weekly** and flag employees exceeding standard hours for three or more consecutive weeks as early warning signals for manager follow-up conversations.
- **Redistribute excess workload** through team restructuring, project reprioritization, or temporary hiring before overtime becomes chronic and structural rather than situational.
- **Require manager approval for sustained overtime** and track overtime rates by role and department as a leading attrition indicator in quarterly HR reviews.

- **Audit workload distribution in Sales and R&D quarterly,** the two largest departments by headcount, where overtime concentration is likely highest.

A reduction in chronic overtime for high-risk employees would directly reduce the largest risk multiplier identified in the model.

4.2 Job Satisfaction Monitoring (JobSatisfaction *OR* = 0.736, *p* < .001)

Each one-point increase in job satisfaction reduces attrition odds by 26.4 percent. The average satisfaction score for employees who left was 2.47 out of 4.

- **Replace annual surveys with quarterly pulse surveys** of three to five questions measuring satisfaction at the team and role level. Annual surveys capture satisfaction too infrequently to enable timely intervention.
- **Set an early intervention threshold at 2.0 or below.** Employees scoring at or below this level on the four-point scale should receive a structured one-on-one manager conversation within 30 days of the survey.
- **Train managers in stay interview techniques** focused on understanding what keeps employees engaged rather than why they might leave. Direct manager relationships are consistently among the most influential retention factors in the research literature (Hom et al., 2017).
- **Track satisfaction trends over time** and evaluate whether HR interventions produce measurable improvement for flagged employees. Two consecutive quarters of declining scores should trigger a formal retention review.

4.3 Early-Tenure Retention Programs (YearsAtCompany *OR* = 0.942, *p* = .007; Age *OR* = 0.954, *p* < .001)

Both tenure and age are significant negative predictors. Employees who left averaged 5.13 years at the company versus 7.37 years for those who stayed. Younger employees also show higher attrition, consistent with early-career mobility patterns.

- **Develop structured 90-day onboarding programs** for all new hires in high-attrition roles including Sales Representatives and Laboratory Technicians. Programs should clarify role expectations, introduce key stakeholders, and include early feedback conversations.
- **Assign peer mentors from tenured employees to new hires.** Mentorship builds organizational embeddedness, which Mitchell et al. (2001) identify as a primary mechanism reducing voluntary turnover beyond job satisfaction.
- **Conduct structured check-ins at 30, 90, and 180 days** with all new employees to identify dissatisfaction or role misalignment before it leads to departure.
- **Create visible career progression pathways** for entry-level roles such as Sales Representative and Laboratory Technician, showing realistic timelines and requirements for advancement to senior positions.

4.4 Work-Life Balance Policies (WorkLifeBalance *OR* = 0.676, *p* = .001)

Each one-unit improvement in work-life balance reduces attrition odds by 32.4 percent. Combined with the overtime finding, this predictor reinforces that organizational demands on personal time are a central and reinforcing driver of departure.

- **Implement flexible scheduling options** including compressed workweeks, hybrid or remote work arrangements, and flexible start and end times where operationally feasible.
- **Establish clear norms around after-hours communication** by setting explicit expectations about email response windows and discouraging non-emergency contact outside business hours.
- **Hold managers accountable for work-life balance scores** within their teams through annual performance evaluations, aligning managerial incentives with organizational retention goals.

4.5 Role-Specific Retention for Sales Representatives and Laboratory Technicians

Sales Representatives ($OR = 9.986$, $p = .023$) and Laboratory Technicians ($OR = 6.172$, $p < .001$) have statistically significant elevated attrition risk in the model compared to Managers, independent of all other predictors. This means the role itself carries elevated risk beyond what satisfaction, overtime, or income explain.

- **Sales Representatives:** Review commission structures and quota targets for achievability. Unrealistic targets in high-pressure environments are a well-documented driver of Sales role attrition. Consider adding a base pay component for quarters where market conditions make quota achievement difficult.
- **Laboratory Technicians:** Evaluate career development pathways. High attrition in this role may reflect limited advancement opportunities or a perceived salary ceiling. Creating a Senior Laboratory Technician designation with higher pay and expanded responsibilities provides a visible retention pathway.
- **Study what makes Manager and Research Director roles retentive.** These roles show 4.9 and 2.5 percent attrition respectively. Identifying the structural factors that contribute to low attrition in these roles and replicating those conditions in high-attrition roles is a high-value strategic exercise.

Conclusion

This project demonstrates how business intelligence and predictive analytics can convert standard HR data into specific, actionable workforce retention insights. Using the IBM HR Analytics Employee Attrition Dataset with 1,470 records and following the CRISP-DM framework, the analysis moved from business problem definition through descriptive analysis, logistic regression modeling, and deployment-ready recommendations. All figures in this report were computed directly from the dataset.

The logistic regression model achieved 86.1 percent accuracy and an AUC of 0.730 on the held-out test set, meeting both pre-defined success thresholds. All three research hypotheses were confirmed. OverTime is the strongest predictor with an odds ratio of 4.195, meaning overtime workers are more than four times as likely to leave. JobSatisfaction, WorkLifeBalance, YearsAtCompany, and Age are all statistically significant predictors. Monthly income, while showing a large descriptive difference between groups, is not significant in the multivariate model after controlling for job role. Sales Representatives and Laboratory Technicians carry significantly elevated role-level attrition risk independent of all other predictors.

The most important practical conclusion from this analysis is that the dominant attrition predictors are organizational conditions within management control. Overtime schedules, job satisfaction, and work-life balance are all outcomes of decisions organizations make actively. That means the findings translate into specific, manageable interventions rather than abstract observations about who leaves.

Future extensions of this work could incorporate ensemble methods such as Random Forest or Gradient Boosting to improve recall, apply the model prospectively to flag current high-risk employees, and test whether recommended interventions produce measurable reductions in attrition over time. This project establishes the analytical foundation on which those more advanced applications could be built.

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